

Research is How the Future is Created

CSEG3060 · Research Methodology in Computer Science

Lecture 0 — Why any of this matters

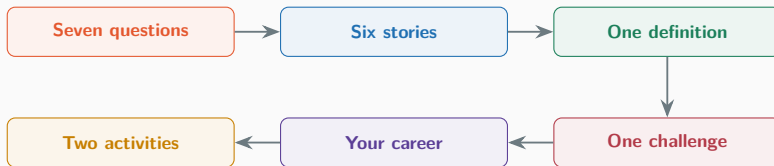
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B.Tech CSE — Final Year



Today is not a syllabus lecture



The only promise I am making

You will leave this room able to name **one problem in the world** that you personally would like to solve.

Seven Questions

Seven questions. Shout the answers.

How was
ChatGPT created?

Why is
Google Search
so accurate?

How does **Netflix**
know what you
will watch next?

How does a **Tesla**
drive itself?

How does **Instagram**
pick your reels?

Who invented
Git — and why?

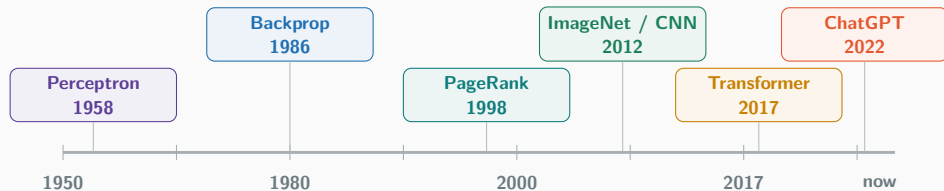
How does **Maps**
predict traffic
you have not hit yet?

**And: why does
UPI never charge
you a rupee?**

Do not answer politely. Answer loudly. There are no wrong answers for the next two minutes.

**None of these
happened by accident.**

Every one of them is the output of years of research



The 64-year sentence

The product you used this morning is the **last two years** of a story whose first **sixty** years were written in research papers by people whose names you do not know.

Your Turn: Think–Pair–Share

Activity 1 · Think – Pair – Share



The question

Which single piece of technology has changed *your* life the most?

Rule: you may not say “the internet” or “my phone”. Be specific.

Whatever you said, a researcher got there first

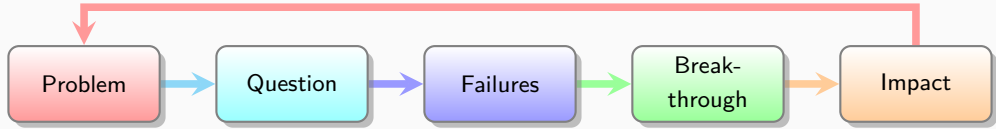
You said	The research question behind it	Field
Google Maps	Can shortest paths be computed on a graph with millions of edges, in milliseconds?	Algorithms, Dijkstra 1959
WhatsApp	Can two strangers agree on a secret key over a channel everyone can read?	Cryptography, D-H 1976
UPI / GPay	Can a payment be final, instant, and free at national scale?	Distributed systems
YouTube / Reels	Can video be compressed 100× without your eye noticing?	Signal processing, codecs
Face Unlock	Can a face be a stable numeric vector under any light?	Computer vision
ChatGPT	Can attention alone replace recurrence in sequence models?	Deep learning, 2017

Notice

Every row is a **question**, not a product.

Six Stories

The anatomy of every research story



The step everyone skips when they tell the story: FAILURES

Watch for this shape in each of the next six slides.

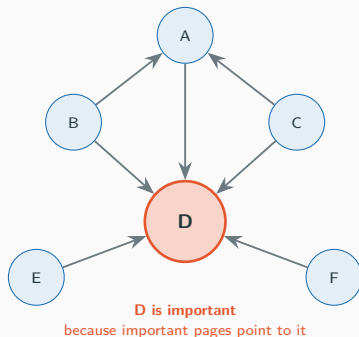
Story 1 · Google — two PhD students and a library problem

Problem (1996). The web had 25 million pages. Search engines ranked by *how many times your word appeared*. Spammers won.

Question. *Can the importance of a page be inferred from who links to it, rather than from what it says?*

The leap. Treat the web as a giant matrix. Importance is its principal eigenvector. Solve by repeated multiplication.

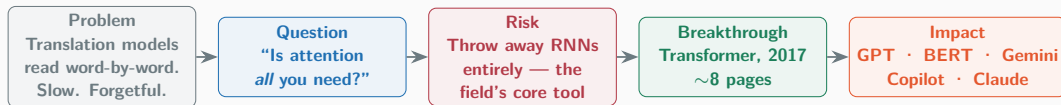
Impact. A Stanford research project, published as a paper, became a company now used ~ 8.5 billion times a day.



Think about it

The maths (Perron–Frobenius, 1907) was **ninety years old**. What was new?

Story 2 · ChatGPT — one paper, eight authors, five years



The whole idea, in one line

Let every word look at every other word *at the same time*, and *learn* how much to care about each.

What students get wrong

"OpenAI invented ChatGPT."
OpenAI *productised* a public research paper written at Google, which anyone — including you — could read for free.

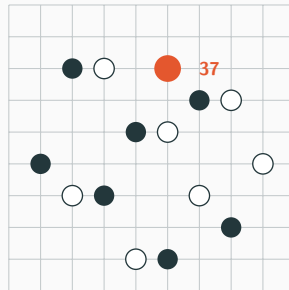
Story 3 · AlphaGo — move 37

Problem. Go has more legal positions than atoms in the observable universe. Brute force is hopeless. Experts said 2016 was “at least a decade away”.

Question. *Can a system learn intuition — a feel for a good position — instead of enumerating moves?*

Method. Policy network + value network + Monte-Carlo tree search + self-play. Millions of games against itself.

Move 37, game 2. A move no human would play. Commentators called it a mistake. It won the game. Lee Sedol left the room.

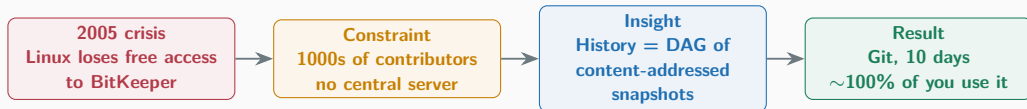


“It’s not a human move. I’ve never seen a human play this move.” — commentator, live

Discuss — 60 seconds

If a system finds a strategy no expert would have chosen, **who made the discovery?**

Story 4 · Git — written in ten days, out of anger



Why this story matters to you

Git was not a lab project. It was a **well-defined problem**, a **hard constraint**, and one person who read enough to know that hashing + DAGs solved it.

Ask yourself

What tool do you use every day that *annoys* you?
Write it in the margin. We come back to it at the end.

Story 5 & 6 · UPI and ISRO — research at national scale

UPI · NPCI

Question. Can a payment be *instant, final, free*, and work between *competing* banks?

Hard parts. Idempotency. Two-phase settlement. Fraud at 10^{10} scale. An addressing scheme humans can type.

Impact. > 15 billion transactions a month; copied by several other countries.

ISRO · Chandrayaan-3

Question. After a failed landing, can we build a lander that survives *its own failure modes*?

Method. Failure-based design: widen the landing zone, add sensor redundancy, simulate thousands of off-nominal descents.

Impact. First soft landing near the lunar south pole — at a cost below several Hollywood films.

Pattern Chandrayaan-2 “failed”. Chandrayaan-3 was **built out of that failure**. In research, a documented failure is an asset.

Why Research Matters

The chain that runs the modern world



Research	Innovation	Technology	Product	Company	Society
Transformer paper	Pre-training at scale	LLMs	ChatGPT	OpenAI	How 800M people write
PageRank paper	Link-based ranking	Web-scale index	Google Search	Alphabet	How the world looks things up
Public-key crypto	Key exchange	TLS/HTTPS	Every “padlock”	Whole internet economy	Online banking exists

Note the direction Nothing flows right-to-left. **You cannot skip step one.**

Seven fields. One question each. All of them are CS.

HEALTHCARE

Can a model read a scan as well as a radiologist — and say when it is unsure?

CYBERSECURITY

Can we detect an attack we have never seen before, without drowning in false alarms?

CLOUD COMPUTING

Can a million machines behave like one, when some of them are always broken?

ARTIFICIAL INTELLIGENCE

Can a system explain *why* it decided, in words a human can check?

SOFTWARE ENGINEERING

Can we predict which module will fail before a single user does?

AUTONOMOUS VEHICLES

Can a car be safe in situations that appear once in a billion kilometres?

CLIMATE SCIENCE

Can we cut a data centre's energy use without cutting its throughput?

YOUR FIELD

Whatever you care about — there is an open question in it. Unit 1 teaches you how to find it.

Every box is somebody's final-year project, somewhere, right now.

So What Is Research?

Vote. Hands up. Is this research?

Case A

A student builds a **college website** using React and MongoDB. It works. It looks good.

Case B

A student designs a **new pruning method** that shrinks a CNN by 60% with under 1% accuracy loss.

Case C

A student changes a scheduler setting and **battery life improves 40%** on their own phone.

Case D

A student re-runs someone else's published experiment and gets a **different result**.

Case E

A student compares 5 existing algorithms on a dataset nobody has **ever tested them on**.

The test: four questions, not one definition

1. NEW

Is the *answer* unknown — to you *and* to the literature?

2. SYSTEMATIC

Did you follow a method someone else could repeat?

3. EVIDENCE

Can you prove it — data, proof, or measurement, not opinion?

4. SHARED

Is it written down so that others can build on it?

	Case	New	Systematic	Evidence	Shared	Verdict
A	College website	no	—	—	—	Engineering, not research
B	New pruning method	yes	yes	yes	yes	Research
C	Battery +40%, one phone	yes	no	n=1	no	Not yet — but one step away
D	Failed replication	yes	yes	yes	yes	Research (and valuable)
E	5 algorithms, new data	yes	yes	yes	yes	Research (empirical)

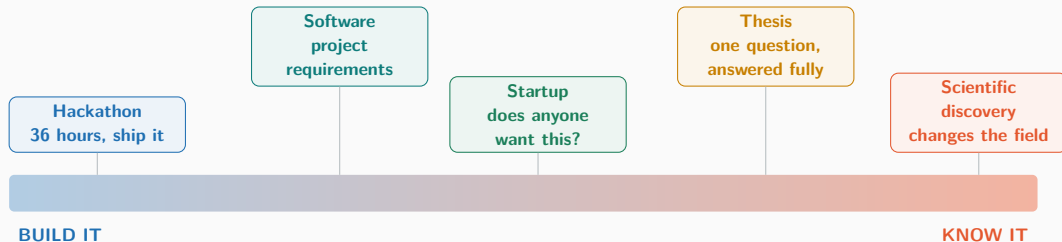
Project vs Research — different goals, different success

	Software Project	Research Project
Starts with	A requirement	A question
Success is	It works, on time	We learned something true
Unknown	<i>How</i> to build it	<i>Whether</i> it can be built at all
Failure means	You failed	You have a result
Output	A working system	A claim + the evidence for it
Audience	Users	Reviewers, then users
Reused by	Whoever installs it	Whoever cites it
Ends when	It ships	Someone else can reproduce it

But they are not enemies

A great final-year project is **both**: a system that works, *and* a question it answers. Build the system to run the experiment.

The same spectrum, five familiar things on it



Two questions for the room

Where does a **Kaggle competition** sit? Where does **your** final-year project sit — and where do you *want* it to sit?

How Research Actually Happens

The research lifecycle — and it is a loop, not a line

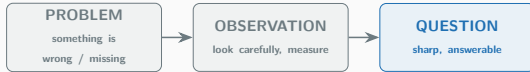
PROBLEM

something is
wrong / missing

The research lifecycle — and it is a loop, not a line



The research lifecycle — and it is a loop, not a line



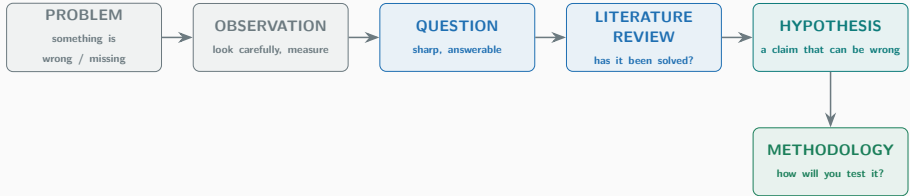
The research lifecycle — and it is a loop, not a line



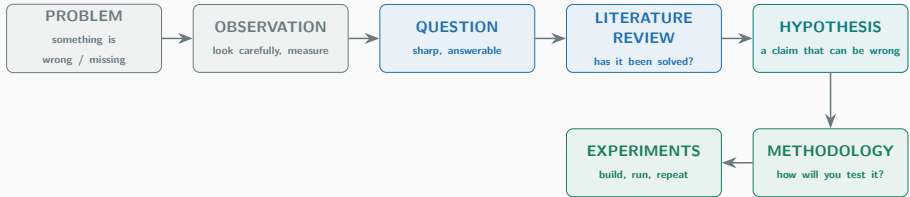
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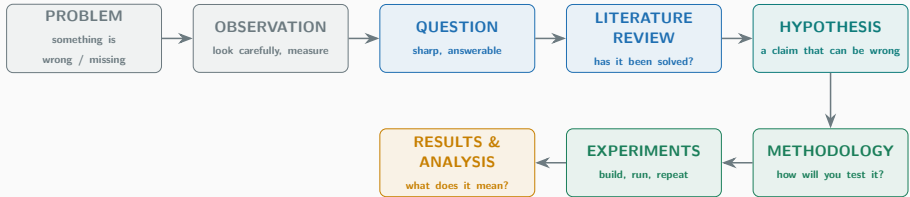
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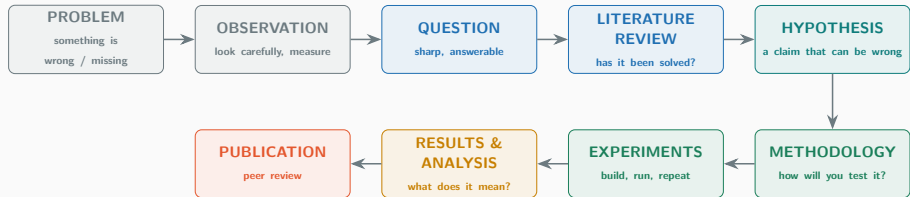
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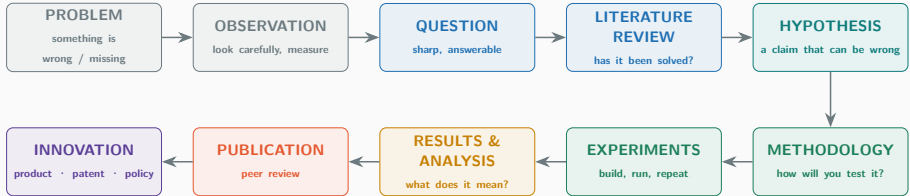
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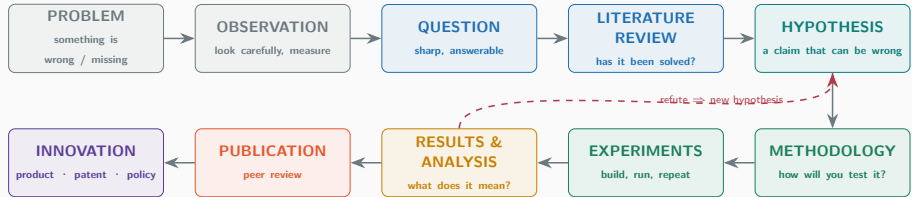
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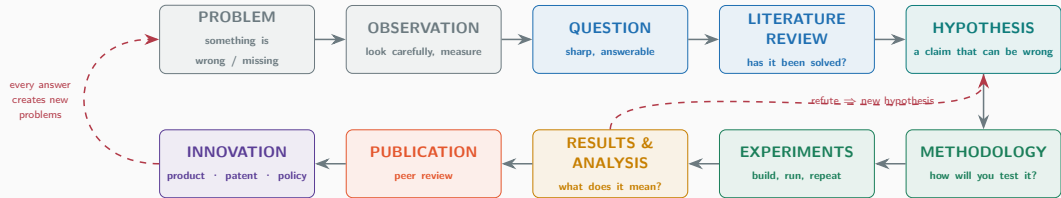
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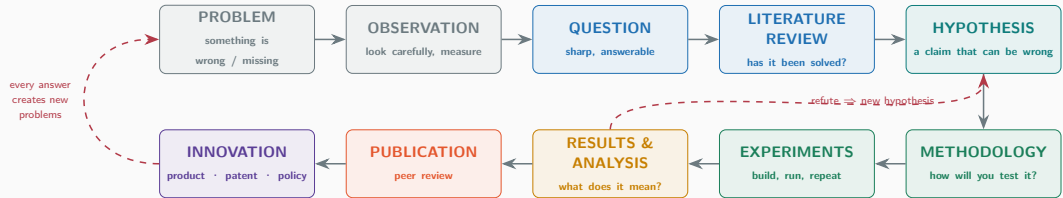
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The research lifecycle — and it is a loop, not a line



The research lifecycle — and it is a loop, not a line



Reality

Steps 6–8 are executed **ten to fifty** times before step 9 happens once.

Why Should *You* Care?

Every door this skill opens

MS / M.Tech abroad

publications > CGPA in most admits

PhD

funded; you get paid to be curious

Research Engineer

Google, MSR, Adobe,
Samsung, ISRO, DRDO

ML Engineer

reads papers weekly, ships models

AI Scientist

invents the method others use

Data Scientist

experiment design, causality, statistics

R&D Engineer

corporate labs, national labs

Software Architect

evaluates options with evidence

Patent Engineer / IP Analyst

Units V-VI are literally this job

Startup Founder

a defensible idea needs a hard question

Product Innovation

decides what is worth building

Faculty

teach the next batch
to ask better questions

Uncomfortable truth

For half of these, a rejected paper on your CV beats a perfect CGPA with nothing behind it.

Even if you never write a paper, you keep five habits



The engineer without these

"It felt faster after my change."
"I couldn't find anything on Google, so I built it from scratch."

The engineer with them

"P95 latency dropped 34% across 3 workloads; here is the profile, and here are the two papers that predicted it."

What This Course Gives You

CSEG3060 — six units, one research lifecycle

UNIT I · Research Problem

9 h

Where problems come from. What makes one *good*. Scope, objectives, data collection, instrumentation.

You leave with: a defensible problem statement.

UNIT II · Research Methods in CS

13 h

Proof, demonstration, empirical study. Deduction vs induction. Modelling, algorithms, performance evaluation.

You leave with: the right method for your claim.

UNIT III · Literature & Ethics

8 h

How to read 100 papers efficiently. Plagiarism. Research ethics and honest reporting.

You leave with: a survey you could publish.

UNIT IV · Technical Writing

5 h

Reports, papers, proposals. Structure, figures, and surviving a review committee.

You leave with: a written research proposal.

UNIT V · IP and Patents

5 h

Patents, designs, copyright. The patenting process. International filing and the PCT route.

You leave with: knowing what you can own.

UNIT VI · Patent Rights & New IPR

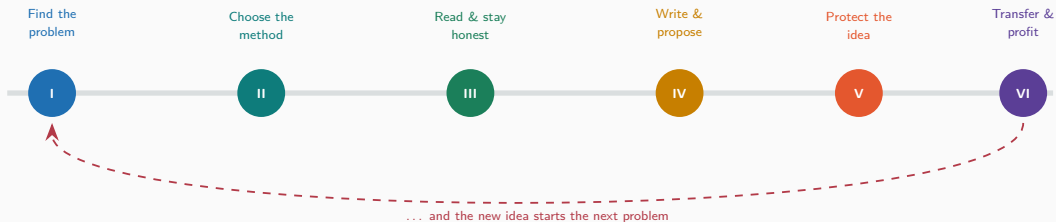
5 h

Licensing and technology transfer. Patent databases. IPR in software. Case studies.

You leave with: turning an idea into an asset.

45 contact hours · IA 50% · Mid-Sem 20% · End-Sem 30%

Where each unit sits in the loop



What I want from you this semester

Bring **your own problem** to Unit I. Every assignment in this course will be applied to *your* problem, not a textbook one.

Your Turn — Two Activities

Activity 2 · Reverse-engineer the research question

Google Translate

Face Unlock

ChatGPT

Spotify
recommendations

WhatsApp

Drone delivery

Self-driving cars

Blockchain

Groups of 4 · 6 minutes · pick TWO

1. What was *impossible* before it existed?
2. Write the research question as “*Can we ... without ... ?*”
3. Name one thing that is *still* broken about it.

Mini Challenge · Three problems, written by you

Take out paper. 5 minutes. Individual.

Write **three real-world problems** you personally wish technology would solve.

At least one must come from something that **annoyed you this week** — remember the Git slide.

Then, next to each one, write

Q: “Can we _____ without _____?”

HEALTHCARE

EDUCATION

ENVIRONMENT

AGRICULTURE

SECURITY

TRANSPORT

ACCESSIBILITY

SOMETHING ELSE

Keep this paper. In Unit I you will turn one of these three lines into a formal problem statement — and possibly into your final-year project.

Close

Five things to take out of this room

1 · Nothing you use is an accident

Every product is the visible end of a long, mostly invisible research chain.

2 · Research starts with a question

Not with a technology, a tool, or a title.
“Can we ... without ...?”

3 · Four tests, not one definition

New · Systematic · Evidence-backed · Shared.

4 · Failure is data, not defeat

A documented negative result is a contribution. Chandrayaan-2 built Chandrayaan-3.

5 · This is a career skill, not an exam topic

Architects, founders, data scientists and product leads all argue with evidence. That is methodology.

Next class

Unit I — and we start from **your** three problems.

**“Research is seeing what everybody
else has seen and thinking what
nobody else has thought.”**

— Albert Szent-Györgyi, Nobel Laureate

So — what problem
will *YOU* solve?

Thank you

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Bring your three problems to the next class.